The Barbero–Immirzi Field as a Hyperbolic Axion–Dilaton: a Fixed Kinetic Mixing, a Validated Spiral Twist Solver, and Predictions Locked Before the Data

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Abstract

Promoting the Barbero–Immirzi (BI) parameter to a field in $\mathcal{N} = 1$ supergravity gives a dilaton–axion pair on a hyperbolic field space of curvature $-2/3$ ($\alpha = 1$), so the BI axion's kinetic term is multiplied by $e^{\lambda\chi}$ with $\lambda = -2\sqrt{2/3}$ fixed (version 1). Here we (i) validate a two-field background solver against the published autonomous system of Ramos-Hamud et al. (agreement 10^{-10}) and implement the model in a Boltzmann code (AxiCLASS), cross-checked against that solver (w to 2×10^{-5}, H to 8×10^{-5}); (ii) show that the fixed geometry is a *weak* kinetic coupling, $\xi = \sqrt{2/3} \simeq 0.82$, while the DESI benefit reported for strong mixing requires $\xi \gtrsim 370$, so in that regime the "Spiral Twist" is plain thawing quintessence ($w_0 \simeq -0.84$, $w_a \simeq -0.24$); (iii) identify the remaining route, the BI angle acting as a fraction f of the dark matter, as dark matter coupled to the dilaton with a fixed strength 0.816 per unit f. It bends the w_0–w_a reading toward DESI in proportion to f, but existing coupled-dark-matter fits cap $f \lesssim 6$–9%, where the bend is $\Delta w_a \simeq -0.05$ to -0.09; one setting reaches the edge of the DESI-like region ($w_0 = -0.80$, $w_a = -0.40$, $f = 9.7\%$). We report solar-system, inflation and early-energy checks, a closure caveat in the coupled AxiCLASS build, and forward predictions (DESI DR3 BAO, $w \geq -1$, S_8, CMB birefringence, tensor ratio r, gravitational-wave parity) hash-locked on 2 November 2026 before those data exist.

1 Status

Exact: the field-space geometry and λ (§2). Validated numerically: the solver and its AxiCLASS implementation (§3). Everything else is a hypothesis with its checks reported, including failures and tensions. In the companion analysis, the minimal late-time Twist field's best data preference (DES Y5 supernovae, $\Delta\chi^2 = -6.23$) is a calibration effect: splitting that sample at $z = 0.1$, neither half prefers the field ($+0.53$ above, -0.29 below; the whole gain is the low/high-redshift offset, and w_0–w_a behaves the same). On Union3 and DES-Dovekie the preference weakens to -1.82 and -1.33.

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2 Geometry

Gates, Ketov & Yunes [1]: $\Omega = e^{\Phi+\bar{\Phi}}(\Phi+\bar{\Phi})$, $K = -3\ln(-\Omega/3)$, $\Phi = \phi + ia$, $\phi < 0$. Then $K_{\Phi\bar{\Phi}} = 3/(\Phi+\bar{\Phi})^2$; with $|\phi| = e^{\sqrt{2/3}\chi}$ and $\tilde{a} = \sqrt{3/2}\,a$,

$$\mathcal{L}_{\rm kin} = -\tfrac{1}{2}(\partial\chi)^2 - \tfrac{1}{2}e^{\lambda\chi}(\partial\tilde{a})^2, \qquad \lambda = -2\sqrt{2/3} = -2\sqrt{2/(3\alpha)}\big|_{\alpha=1}. \tag{1}$$

In the notation of Ramos-Hamud et al. [2] ($W = e^{-\xi\phi_1}$) this is $\xi = -\lambda/2 = \sqrt{2/3}$.

3 Validation

Solver. Background equations for χ and θ (the BI angle) with matter, radiation and closure. Limits: exact ΛCDM; single field; null energy condition (true $w \geq -1$ always). Literature mirror: the late-time attractors of the autonomous system of [2] (exponential potential, kinetic coupling) for $(\lambda_p, \xi) = (2,10), (2,100), (8,20)$: $w = -0.818182, -0.980198, -0.666667$ in both codes (differences $\leq 5\times10^{-10}$), with matching energy split. Our attempt to reproduce the Fig. 1 benchmark of Toomey et al. [3] did not succeed (0 of 86 settings of the inputs that paper does not state); we therefore rely on the [2] mirror.

Ghosts. The kinetic matrix $\mathrm{diag}(1, e^{\lambda\chi})$ is positive definite and both sound speeds are 1: no ghost, gradient instability or strong coupling.

Boltzmann code. In the coupled AxiCLASS build, an exponential field plus dark matter coupled as $m_c \propto e^{\beta\phi}$ agrees with the solver to 3×10^{-8} ($\beta = 0$) and 2×10^{-5} in w, 8×10^{-5} in H ($\beta = 0.05$).

A closure caveat (found here). For $\beta \neq 0$ that build closes the universe on its *actual* $H(z{=}0)$, which differs from the input h: $H(0)/H_0^{\rm in} = 0.998$ ($\beta = +0.05$), 0.994 ($\beta = -0.05$). Distances and spectra are self-consistent, but quantities labelled with the input h (the reported H_0; σ_8 at $8\,h^{-1}$Mpc) are offset by up to 0.6% in coupled runs.

4 What the fixed geometry implies

Weak mixing. $\xi \simeq 0.82$, whereas the DESI fits of [2] benefit only for $\log_{10}\xi > 2.5$ ($\xi \gtrsim 370$). Reaching that with the hyperbolic geometry would need $\alpha < 5\times10^{-6}$, abandoning the BI derivation. With the decay constant fixed by the Nieh–Yan period ($F = (4/\pi)\sqrt{3/2}\,M_{\rm Pl}$), a scan of 80 settings gives at best $w_0 \simeq -0.84$, $w_a \simeq -0.24$: the thawing relation $w_a \simeq -1.53(1+w_0)$.

Matter-like angle. If the angle oscillates ($m \gg H_0$), its effective mass $m\,e^{-\lambda\chi/2}$ depends on the dilaton: it is dark matter coupled with $d\ln m/d\chi = 0.816$. As a fraction f of the matter this is an effective coupling $\beta_{\rm eff} \simeq 0.816 f$.

κ	f	$\beta_{\rm eff}$	w_0	w_a	thawing w_a at that w_0
1.0	4.0%	0.033	−0.850	−0.256	−0.230
1.2	4.2%	0.034	−0.777	−0.356	−0.341
1.2	9.7%	0.079	−0.795	−0.401	−0.314
1.2	18.1%	0.148	−0.830	−0.599	−0.260

The direction is toward DESI. Two existing results cap f: coupled thawing quintessence gives $|\beta| < 0.073$ [4] ($f \lesssim 9\%$), and the companion Coupled Twist fit prefers $|\beta| \simeq 0.05$ ($f \simeq 6\%$). At those values the bend is $\Delta w_a \simeq -0.05$ to -0.09. DESI's central point ($w_0 \simeq -0.75$, $w_a \simeq -0.86$) is not reached. A full fit (implemented, §3) is the next step.

5 Other checks (rules fixed before computing)

Check	Result	Verdict
Solar system (matter on the supergravity frame)	GR point $\phi = -\frac{1}{2}$ is an attractor; Cassini band $\lvert\Delta\chi\rvert < 0.0072\, M_{\rm Pl}$	passes only if pinned
Simplest potential (constant W)	turning point exactly at $\phi = -\frac{1}{2}$; negative energy	needs uplift
Inflation from $\alpha = 1$	$n_s = 0.9667$ at $N = 60$ vs 0.974 ± 0.003	tension (2.4σ)
Dilaton as early dark energy	$\leq 2.5\%$ at $z = 3500$; G changes $\times 13$ at BBN	too small

Matter blindness (needed if the dilaton rolls) is natural in scale-invariant Einstein–Cartan gravity [5, 6]; whether it coexists with $\alpha = 1$ is open.

6 Predictions locked before the data (2 November 2026, 16:19:51 CST)

The attached prediction file `PREDICTIONS_LOCKED_2026-11-02.md` (SHA-256 begins `b629f534`, ends `23cb`) and the numbers file `forward_predictions.json` freeze, from the stored best fits:

- DESI DR3 BAO: D_M/r_d, D_H/r_d at the DR2 redshifts for the Late Twist, Coupled Twist, ΛCDM and w_0–w_a; the rule is $\Delta\chi^2 \leq -4$ preferred, $\geq +4$ disfavoured.
- $w(z) \geq -1$ at all z for the minimal and Coupled Twist: a $> 3\sigma$ model-independent phantom bin excludes them.
- S_8: the Late Twist is indistinguishable from ΛCDM; the Coupled Twist is higher (0.828).
- CMB birefringence ratio $R = 0.997$ (if photon-coupled).
- Tensor ratio from $\alpha = 1$ inflation: $r = 0.0033$–0.0048; $r < 0.002$ (95%) excludes it.
- Gravitational-wave velocity birefringence $\sim 10^{-43}$–$10^{-42}\,$GeV; a detection above $10^{-41}\,$GeV is not the Twist.

7 How to falsify

(1) A full fit of the effective Spiral Twist (κ, $\beta_{\rm eff}$; no DES Y5) that does no better than thawing. (2) A derivation showing the dilaton must couple to matter (Cassini then excludes a rolling dilaton). (3) Any locked prediction above failing on its named data.

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References

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